

Exercise 6 – Classification

Report:

In this exercise, I used two machine learning models (classifiers) to predict if a person earns more or less than \$50,000 a year. The Decision Tree model works by splitting the data into subsets based on the features which create a tree structure of decisions. The Gaussian Naive Bayes model works by calculating the probability of each class from the feature values and selects the class with the highest probability. I chose to use GaussianNB because the dataset has continuous features, meaning they are not bounded by any fixed interval. I used preprocessing techniques like one-hot encoding for turning categorical columns into binary numbers and I also used standard scaler to standardize the numerical columns. In order to train and test the Decision Tree model and the Gaussian Naive Bayes model accurately, I had to split the dataset for training and testing. I split the dataset into 80% for training and 20% for testing. The training set was used to fit the models and the testing set was used to test their accuracy.

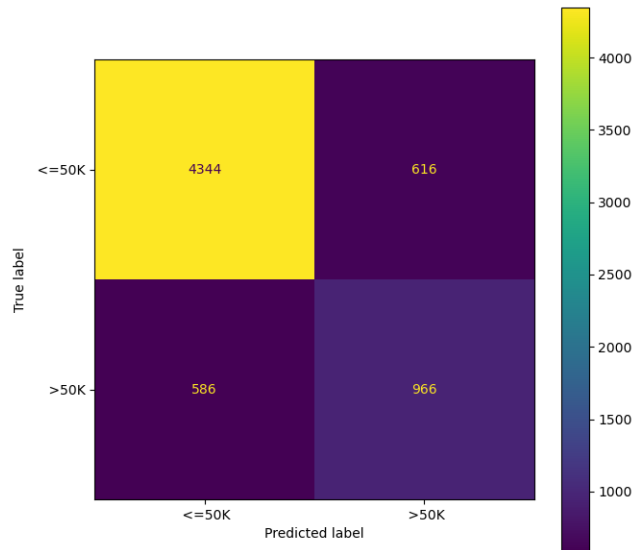
The Decision Tree model resulted in an accuracy score of 81.5%. The confusion matrix showed 4344 true negatives, which means that the model correctly predicted 4344 cases for the $\leq 50K$ class. It also resulted in 966 true positives, which means that there were correct predictions for the $> 50K$ class. The model had 616 false positives for the $\leq 50K$ class which were incorrectly classified as the $> 50K$ class. Also, there were 586 false negatives which means that there were 586 cases for the $> 50K$ class that were incorrectly predicted as the $\leq 50K$ class.

The Gaussian Naive Bayes model achieved a lower accuracy score of 61.8%. The confusion matrix showed 2592 true negatives, which means that the model correctly predicted 2592 cases for the $\leq 50K$ class. It also resulted in 1436 true positives, which means that there were 1436 correct predictions for the $> 50K$ class. The model had 2368 false positives for the $\leq 50K$ class which were incorrectly classified as the $> 50K$ class. There were 116 false negatives which means that there were 116 cases for the $> 50K$ class that were incorrectly predicted as the $\leq 50K$ class.

Overall, the Decision Tree performed better than the Gaussian Naive Bayes Model. It scored a higher accuracy and also resulted in a more balanced performance across both income classes. The Decision Tree model is able to handle non-linear relationships well because its goal is to achieve a decision boundary that can accurately fit the data points. The Gaussian Naive Bayes model is sensitive to feature scaling (standardization) and performs worse

on complex datasets. The Decision Tree model is better for this kind of prediction task.

Decision Tree confusion matrix:



Gaussian Naive Bayes confusion matrix:

